



INSTITUTO POLITECNICO NACIONAL

Escuela Superior de Computo

Instrumentation and control

Use of Resistive Sensors

Teacher: Hernández Aguilar Jose Luis

Students:

Borbolla Díaz Frida Gabrielle

Gutiérrez Espinosa Jordan Gabriel

Group: 5CV1



Contenido

Objective.	3
Material and equipment used possible of the variable in question.	3
Theoretical introduction.	3
Development of the practice.	6
Questionary.....	7
Observations.	8
Conclusion.	8
Bibliography.	8



Objective.

The student will learn about to use resistive sensors, as well as to calibrate the different components of a linear measurement system, in order to find the voltage value corresponding to the variable under measurement.

Material and equipment used possible of the variable in question.

Equipment provided by the laboratory's personal: the Material to be brought by the students:

2 Multimeters

1 Variable VCD source

8 BCN Tips — Cayman

2 BNC Tip - BNC

Protoboard

1 10K Ohm Potentiometers of precision

1 resistor of 1k Ohm

2 resistors of 2k Ohm

3 LM741

Theoretical introduction.

A sensor is an element or device capable of reacting to changes in a given variable, usually transforming them into changes in a second variable, which is easier to interpret as a measurement value.

In these cases, we are particularly interested in sensors that can "translate" a certain variable to be measured in an electrical variable (voltage, resistance or current).

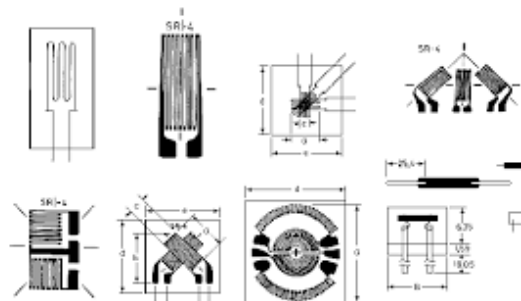
There are several classifications according to different criteria such as:

- The variable they measure (temperature, pressure, humidity, distance, speed, etc.).
- The nature of the output variable (digital and analog).
- Type of variable you have at the output (resistive, capacitive and inductive).

The last criterion is useful because depending on the output variable, the way of use of the sensor and the conditioning that may be required varies.

Resistive sensors.

They are those where the variable to be measured produces a change in the resistance of the element or material used as a sensor. On the basis that the resistance of some materials is easily altered by some external factors (temperature, light capacity or deformation of exposure to the material).





Potentiometric sensors.

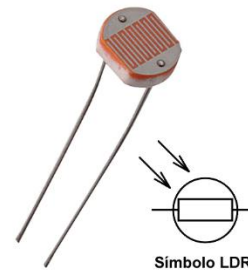
They are those where the resistance of the element varies with a physical movement, the most common being the rotation potentiometers in which the resistance between the central terminal and the lateral ones varies according to the angle of rotation of the knob.

Moving the cursor changes the length of material. between terminals and therefore resistance varies, the main drawbacks are: friction and mechanical movement.



Photoresists.

Also known as photoresistors are elements in which resistance decreases when light hits the element. They are made up of semiconductors and their operation is based on the photoelectric effect, the common elements that compose it are thallium sulfide, cadmium sulfide, lead sulfide and cadmium selenide.



Thermistors and RDT.

Thermistors are built based on semiconductors and are presented in two different types, depending on the characteristics of the material are constituted by:

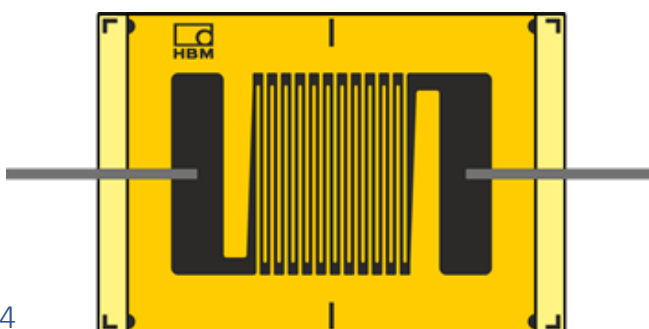
- NTC thermistors: These are those that have the negative temperature coefficient, that is, in which an increase in temperature results in a decrease in resistance.
- PTC thermistors: These are those that have a positive temperature coefficient, that is, in which an increase in temperature results in an increase in resistance.

Its lack of linearity and self-heating problems are some of the main disadvantages, while its advantages are low cost, high sensitivity and rapid response.



Strain gauges.

They are sensors based on the piezoresistive effect, that is, the property that some materials have to change their resistance when subjected to mechanical stress. These elements are used to measure strain, pressure or load.





Capacitive sensors.

They are those where the variable to be measured is reflected in a change in the capacity of an element, either by an alteration of the dielectric constant or in the structure of the element, the variables that can be measured by this type of sensors are motion, electric fields, chemical compositions, depth, tactile sensors and accelerometers.



Inductive sensors.

They are those where the variable to be measured produces a change over the inductance of the sensor (a coil), are used as proximity sensors for metal objects. Its function is based on an electromagnetic induction principle, when a metal object comes into contact with the alternating magnetic field generated by the coil eddy currents are induced in the object.



Piezoelectric sensors.

It makes these materials can be used as sensors that transform pressure into a small difference in electrical potential, piezoelectric elements are used in pressure, acceleration and force sensors.



Thermocouples.

They're temperature sensors that base their operation on the seebeck effect, when joining two wires made of different metals and applying a temperature change at the point of union creates a small potential difference between the two terminals which is proportional to the temperature change.



Development of the practice.

1.- Resistance bridge.

Arme the circuit of Figure 1 considering that the Wheatstone bridge must remain in equilibrium, that is $V_A - V_B = 0V$, when the Precision R1 potentiometer has a value of $1K\Omega$. To do this, determine the value of the resistors R2, R3 and R4 according to what is seen in class and considering that R1 will change from $1K\Omega$ to $10K\Omega$

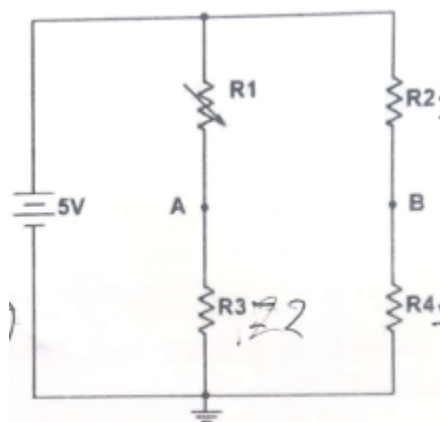


Figure 1.

Perform the calibration of the resistive bridge according to the resistance variation in R1 and setting the $V_{AB} (MAX) = 2.5V$

2.- Amplification of the signal

Once the resistance bridge has been calibrated, proceed to connect it to the circuit of Figure 2, in which you must use the LM741 energized with 7V. Adjust the circuit gain in Figure 2 so that your V_{SAL} goes from zero to 5V in the V_{AB} range set for the resistive bridge.

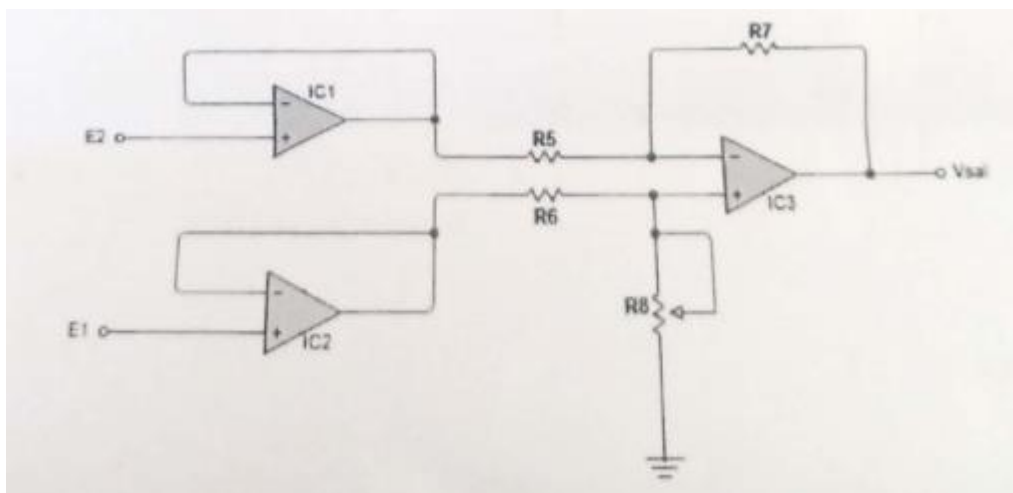


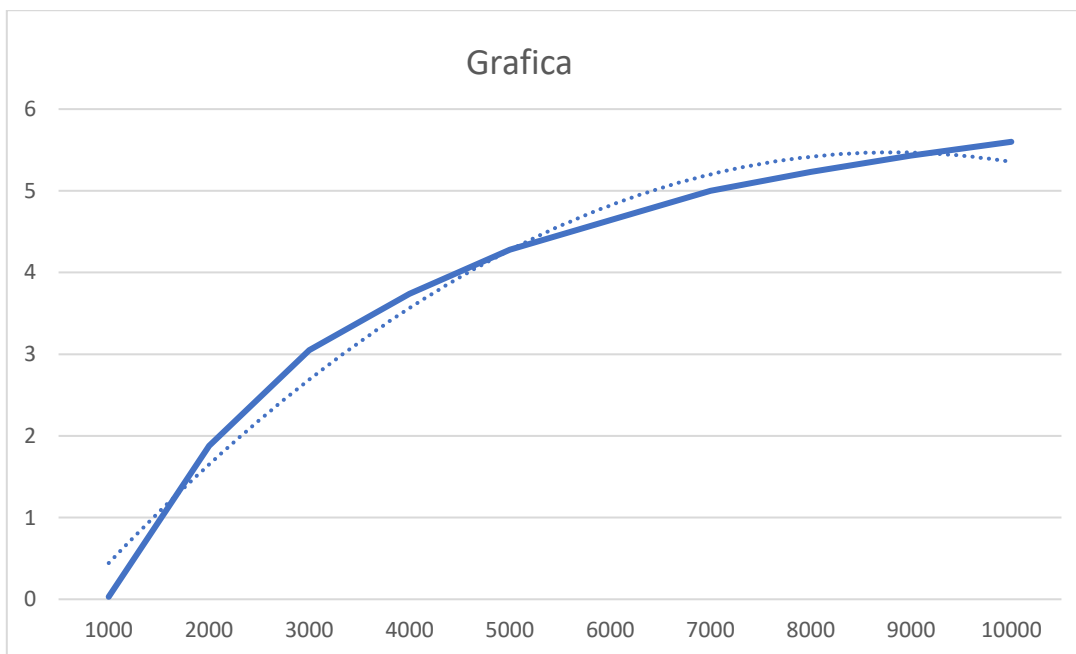
Figure 2



Set the required resistance values for R5, R6, R7, and R8. Then proceed to fill the table 1 adjusting the value of R1 each time with the ohmmeter without it being energized and measuring the voltages that in this table are requested with the DC voltmeter after energizing the circuit.

Resistor R1 (Ω)	V_B (Volts) measured	V_A (Volts) measured	$V_A - V_B$ (Volts) measured	V_{SAL} (Volts) measured	Resistance in R1 (Ω) Calculated
1000	3.30	3.37	0.07	0.03	0.96
2000		2.52	-0.78	1.88	1.96
3000		1.99	-1.31	3.05	3.02
4000		1.57	-1.73	3.74	4.36
5000		1.43	-1.43	4.28	4.99
6000		1.25	-2.05	4.64	6.01
7000		1.118	-2.182	5.00	6.94
8000		1.01	-2.29	5.23	7.9
9000		0.91	-2.39	5.43	8.98
10000		0.841	-2.459	5.60	9.89

Then make the V_{SAL} graph. against Resistance in R1 and get the equation for that graph.



$$V_{SAL} = 2.4288 \ln(x) - 16.558 \text{ (logarithmic method)}$$

Questionary.

1.- What variable is measured with the circuits implemented in this practice? Electrical resistance and voltage are measured.



2.- With sensitivity set in mV/Ω , after signal conditioning and if the $V_{\text{ref}}=5\text{V}$. How many bits should the ADC be required for this measurement? 6 bits.

3.- Is the answer found linear? Why? No, because the phase response of the filter is not linear in the frequency range, that is, it is not directly proportional to the frequency.

4.- Investigate and indicate another way to perform the signal conditioning performed by the circuit in Figure 2. They are passed through different filters or analog processors, converting those signals from analog to digital and then processing the data with a microcontroller and acting through logical outputs of the processor or through a digital to analog converter.

5.- Investigate, which resistive transducer is recommended for measuring distance? Capacitive transducers pick up signals from distance and pressure exerted by one plate on the other.

Observations.

In signal amplification, the resistors R5 and R6 must be directly double.

Conclusion.

In the course of the practice, we learned 3 things, the use of the precision potentiometer who helps us to have a better results when we need a variance in ours resistances in our circuits, The Wheatstone bridge and how it works for resistance calculation by comparison with other known resistance and also the use of the LM741 as amplifier for the operations of our circuits and how these should be used in future practices.

Bibliography.

<https://ntcsensors.com/Qu es el sensor resistivo />

<https://tableroalparque.weebly.com/uploads/5/1/6/9/51696511/introducci%C3%93n a los sensores.pdf>

<https://transductor.net/transductor-capacitivo/>